FSL-Unet: Full-Scale Linked Unet With Spatial–Spectral Joint Perceptual Attention for Hyperspectral and Multispectral Image Fusion

Xianghai Wang, Xinying Wang, Keyun Zhao, Xiaoyang Zhao, and Chuanming Song

***Abstract*— The application of hyperspectral image (HSI) is more and more extensive, but the lower spatial resolution seriously affects its application effect. Using low-resolution HSI (LR-HSI) and high-resolution (HR) multispectral image (MSI) fusion technology to achieve super-resolution reconstruction of HSI has become a mainstream method. However, most of the existing fusion methods do not make full use of the large-scale range of remote sensing images and neglect the preservation of spatial–spectral information in the fusion process. Considering that the spectral information in fused HR-HSI mainly depends on HSI, and the spatial information mainly depends on MSI, this article proposes a full-scale linked Unet with spatial–spectral joint perceptual attention (SSJPA) for hyperspectral and MSI fusion (FSL-Unet). The FSL-Unet consists of two modules. The first is the spatial–spectral attention extraction (SSAE) module, which is used to calculate the spectral attention of LR-HSI and the spatial attention of HR-MSI at different scales. The second is the full-scale link U-shaped fusion (FLUF) module, which adopts a multilevel feature extraction strategy, using denser full-scale skip connections to explore feature information in a finer-grained range, enabling the flexible combination of multiscale and multipath features. At the same time, we propose SSJPA on the encoder side of FLUF. SSJPA can make full use of the attention maps computed by the SSAE and then effectively embed spatial and spectral information into the fused image, enabling uninterrupted information transfer and aggregation. To demonstrate the effectiveness of FSL-Unet, we selected five public hyperspectral datasets for experiments. Compared with the other eight state-of-the-art fusion methods, the experimental results show that the FSL-Unet achieves competitive results. The source code for FSL-Unet can be downloaded from https://github.com/wxy11-27/FSL-Unet.**

***Index Terms*— Hyperspectral image (HSI), image fusion, multispectral image (MSI), perceptual attention, Unet.**

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I. Introduction

THE emergence of imaging spectrometers makes it possible to obtain hyperspectral images (HSIs) with continuous spectral information. In the imaging process, the hyperspectral imaging spectrometer simultaneously images the ground objects in dozens or hundreds of bands, which can obtain continuous spectral information of the ground objects and realize the simultaneous extraction of the spatial information, radiation information, and spectral information of the ground objects. Due to its high spectral resolution, HSI can provide more abundant ground information, so it has great application value in the fields of change detection [1], [2], target detection [3], [4], [5], land cover classification [6], [7], [8], and so on. However, a higher spectral resolution will inevitably affect the spatial resolution of HSI. In fact, the original design of the imaging spectrometer is to compromise the parameters, such as spatial resolution, spectral resolution, and radiation sensitivity, so that the acquired HSIs have a lower spatial resolution. Based on this, a method of enhancing low-resolution HSI (LR-HSI) to achieve super-resolution by software methods has been proposed and has become a frontier direction in the field of remote sensing image intelligent processing [9]. Contrary to HSI, conventional MSI has a smaller spectral resolution and larger spatial resolution. Because HSI and MSI have certain complementarities, the method of obtaining high-resolution HSI (HR-HSI) by fusing LR-HSI and HR-MSI has attracted more and more attention.

In recent years, more and more scholars have carried out research on the fusion task of LR-HSI and HR-MSI. Based on their respective characteristics, these fusion methods can be classified into four categories: the pan-sharpening method (pan-sharpening-based method), the matrix factorization method (MF-based method), the method based on tensor representation (TR-based method), and the method based on convolutional neural network (CNN-based method) [10].

Pan-sharpening methods were originally proposed to improve the spatial resolution of MSIs by fusing panchromatic images (PANs) and multispectral images (MSIs). It can be divided into two categories: component substitution (CS) [11], [12] and multiresolution analysis (MRA) [13]. With the popularization of hyperspectral remote sensing systems, scholars have gradually applied pan-sharpening to the fusion task of PANs and HSIs [14], [15], [16]. However, because the spectral resolution of the PAN is too small, the problem

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of spectral distortion is easy to occur in the fusion process. To solve this problem, more and more people adopt the pan-sharpening method for HSI and MSI fusion. In order to better apply the pan-sharpening method to the fusion task of HSI and MSI, Zhao *et al.* [17] proposed a new framework, which first divided the bands of HSI into several groups, and then used pan-sharpening methods to fuse the MSI and HSI of each group. In terms of theory, Selva *et al.* [18] proposed a new framework named hypersharpening and proved that hypersharpening has fusion results better than existing pan-sharpening methods. The method based on pan-sharpening has the advantages of low computational cost and fast calculation speed, but there are many problems when the pan-sharpening algorithm is directly applied to the fusion of HSI and MSI. For example, when the spectral resolution of HSI and MSI is quite different, there will be severe spectral distortion in the fusion results.

The MF-based fusion method first decomposes the 3-D HSI into a 2-D matrix and then transforms the fusion problem into a calculation problem of coefficients and spectral bases. MF-based methods include low-rank methods [19] and sparse representation methods [20], [21]. Due to the low spatial resolution of HSIs, a large number of mixed pixels exist, and unmixing technology [22], [23], [24] has been developed in recent years. Kawakami *et al.* [25] proposed a sparse MF (SMF) fusion of HSI-MSI, which applies the unmixing algorithm to the hyperspectral input to achieve the purpose of spectral preservation. Yokoya *et al.* [26] designed a coupled nonnegative matrix factorization (CNMF) unmixing method to solve the fusion problem of coupled matrix factorization based on a fixed dictionary. Although scholars have found that the MF-based method performs better than the pan-sharpening-based fusion method during the experiment, it needs to constantly solve complex optimization problems. In addition, the MF-based method is very computationally intensive, and its parameters are not uniform and difficult to set.

TR-based fusion methods have been an active topic, including Tucker decomposition [27], canonical polyadic (CP) decomposition [28], tensor singular value decomposition (t-SVD) [29], and tensor-train decomposition [30]. Dian *et al.* [31] proposed a novel algorithm, called NLSTF, for HSI super-resolution based on nonlocal sparse tensor decomposition. It is believed that the method of nonlocal sparse tensor decomposition can make full use of the HSI spatial–spectral structure and the nonlocal spatial self-similarity of HSI. Next, Dian and Li [32] proposed a novel subspace-based low-tensor multirank regularization method, which first learns spectral subspaces from LR-HSI via singular value decomposition and then estimates coefficients via low-tensor multirank priors. Although TR-based methods have achieved good results on some simulated datasets, they still suffer from computationally expensive problems compared to pan-sharpening-based methods.

In recent years, thanks to the rapid development in the field of artificial intelligence, more and more fusion methods based on CNN have appeared. Yang *et al.* [33] designed an architecture named PanNet by focusing on the spectral and spatial preservation problems in the pan-sharpening method. Spectral preservation is better accomplished by propagating spectral information directly into the fused image. Yuan *et al.* [34] proposed a new network for pan-sharpening, which acquires multiscale and multidepth features by introducing multiscale feature extraction and residual learning in CNN. Considering that existing methods ignore the complementary information from HSI and HR RGB images, Han *et al.* [35] proposed two novel CNN architectures, named SSFCNN and ConSFFCNN, respectively, which achieve promising fusion results. Liu *et al.* [36] proposed a two-stream fusion network (TFNet) to fuse PAN and MSI with a feature-level fusion approach and achieved competitive results. Zhang *et al.* [37] proposed an interpretable spatial–spectral reconstruction network (SSR-NET) by designing a new loss function to constrain the spatial and spectral modes. Compared with other methods, CNN-based methods do not rely excessively on hand-designed prior information, and nonlinear functions and convolutional layers can enable CNNs to have a powerful ability to learn image features and achieve precise fusion. However, the generalization problem of CNN-based fusion networks is a big challenge.

To sum up, most of the existing HSI and MSI fusion methods ignore the hierarchical information and cannot preserve spectral information in HSI and the spatial information in MSI well. It is difficult to continuously transfer the spectral information of HSI and the spatial information of MSI to the fused image during the fusion process, which is crucial to the quality of the fusion result. To address the above problems, we propose a novel network Unet with spatial–spectral joint perceptual attention (SSJPA) for hyperspectral and MSI fusion (FSL-Unet) for super-resolution reconstruction of HSI. We believe that most of the spectral information of the fused images should form HSI, and most of the spatial information should form MSI. Therefore, FSL-Unet adopts a pixel-level fusion method, focusing on the multiscale information extraction of the fused image and the information transfer between the fused image, and the original HSI and MSI. More concretely, FSL-Unet consists of two modules: 1) spatial–spectral attention extraction (SSAE) module and 2) full-scale link U-shaped fusion (FLUF) module. First, spectral attention and spatial attention are calculated, respectively, for HSI and MSI in the SSAE module, which is used to enhance the key information in the fused image in the next stage. FLUF is a U-shaped structure, which consists of three parts: encoder, decoder, and skip connections. The FLUF module adopts a pixel-level fusion strategy, fuses the input HSI and MSI to obtain the initial fusion feature map, and extracts finer-grained features through cascade downsampling at the full scale. By adding a denser link between the encoder and the decoder, the multipath information acquisition at the decoder side is realized. It is worth mentioning that we propose the application of SSJPA on the decoder side to keep the uninterrupted information transfer between the fused image and the original HSI and MSI, which can continuously highlight key information and suppress unnecessary information. Finally,

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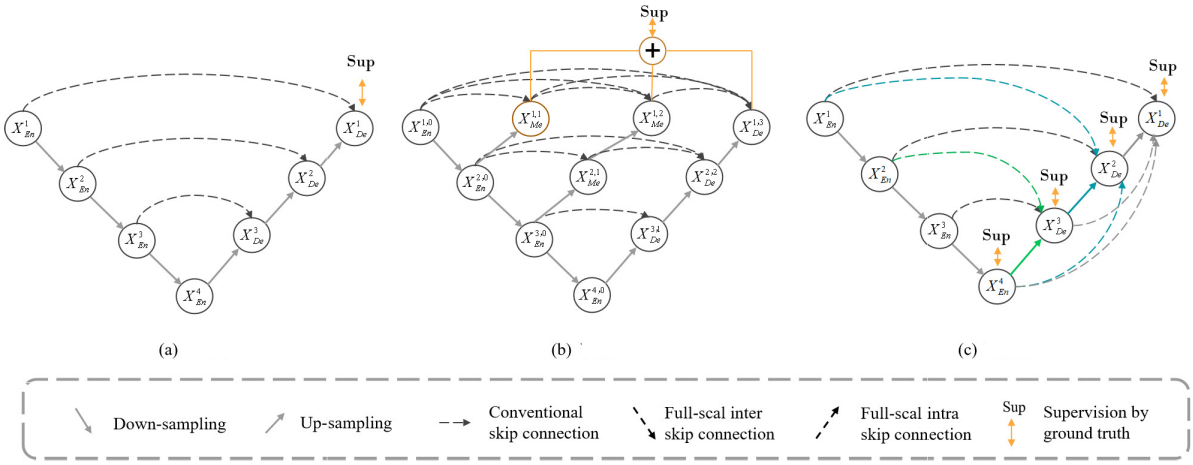


Fig. 1. Architecture diagram of the Unet family, where (a) is U-Net, (b) is Uet++, and (c) is Unet 3+.

the decoder side reconstructs an HR-HSI through cascaded upsampling. The main contributions of this work are reflected in the following aspects.

1) By fully considering the complementarity between HSI and MSI, we propose SSJPA. SSJPA can effectively integrate the spatial information of MSI and spectral information of HSI into the fusion image and realize the preservation of spatial–spectral information during the fusion process.
2) We propose a novel HSI super-resolution reconstruction framework named FSL-Unet. The dense full-scale skip links in FSL-Unet can extract features in the full-scale range, realize a flexible combination of multiscale and multipath features, and greatly improve the quality of fused images.

The remaining chapters of this article are organized as follows. Section II introduces related work, mainly including the Unet family and attention mechanism. Section III describes the proposed FSL-Unet model in detail. Section IV analyzes the experimental results of comparison with the eight most advanced algorithms on five hyperspectral datasets and some ablation experimental results in detail. Section V is the conclusion part.

II. Related Work

In this section, we first provide a general overview of the Unet family. Then, the attention mechanism in deep learning will be introduced.

A. U-Net Family

U-net [38] was originally a well-known network structure for biomedical image segmentation proposed by Ronneberger *et al.* The network structure is mainly divided into three parts, namely, encoder, decoder, and skip connections. As shown in Fig. 1(a), the U-Net network structure presents a U-shape as its name, and its left half is the encoder, which is the downsampling stage, while the right half is the decoder, that is, the upsampling stage. The dotted arrow in the middle is a skip connection, which can splice shallow features and deep features to achieve the fusion of deep and shallow features. Because the shallow layer can capture some simple features, such as color and boundary, and the deep layer can capture some abstract features, the skip connection in U-net can combine deep features and shallow features well to refine the image, and finally, predictive segmentation is performed based on the obtained feature maps. After the last decoding, a 1×1 convolutional layer is used to obtain the final segmentation result.

Zhou *et al.* [39] optimized the above U-Net and proposed an architecture Unet++ based on nested and dense skip connections. As shown in Fig. 1(b), the encoder of Unet++ has the same structure as U-net, and the lower layer features are obtained by downsampling the upper layer features. The main difference between U-net and Unet++ lies in the redesigned skip pathways. Through more dense jump links, the features of the next layer can be upsampled, fused with the features of the previous layer, and then fused with the encoder module of the same layer. This enables the decoder to flexibly fuse features of different semantic scales into the decoder subnetwork and achieve effective capture of features at different levels.

Following U-net and Unet++, Huang *et al.* [40] proposed a new U-shaped architecture Unet 3+ using full-scale skip connections and deep supervision mechanisms. The full-scale skip connection realizes the effective combination of high-level semantic and low-level detail features from different scale feature maps, while deep supervision can learn hierarchical representations from multiscale aggregated feature maps. The Unet 3+ structure makes good use of full-scale feature maps and effectively improves the segmentation accuracy of medical image targets while improving the computational efficiency of the network with fewer network parameters.

It can be seen that the U-net family, in recent years, as a new type of deep learning network architecture with encoder and decoder, provides an important way to effectively mine and fuse multiscale features of images. It can also be seen from its development process that the success of the U-net family largely depends on how to mine features at different scales and efficiently fuse and represent them. The multiencoder and decoder states of the U-Net family make it possible to further study multipath acquisition and fusion of features in different scales, and also reserve room for improvement for further design of more effective network structures.

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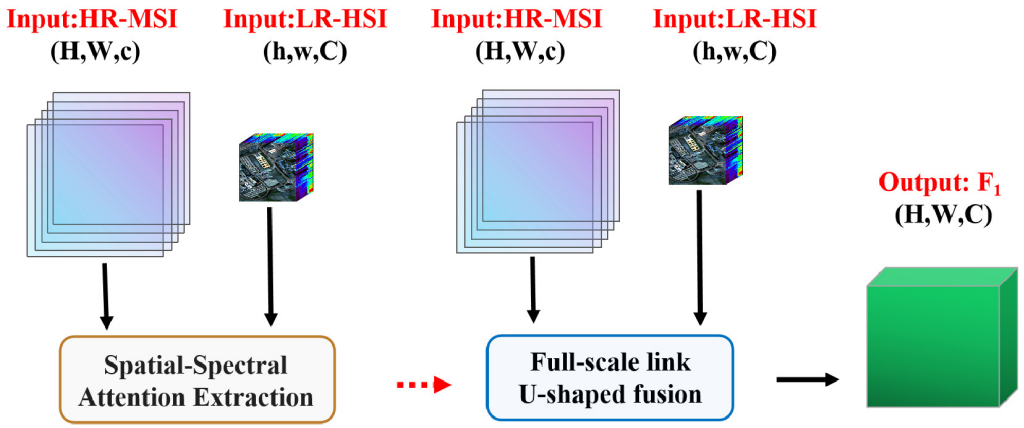


Fig. 2. Overall structure diagram of the proposed FSL-Unet. FSL-Unet consists of two modules, including the SSAE module and the FLUF module. The attention maps calculated by the SSAE module are applied to the FLUF module.

B. Attention Mechanism

In recent years, the attention mechanism has obviously become one of the most worthy of attention and in-depth research in the core technology of deep learning and is widely used in various deep learning tasks, such as speech recognition, image recognition, and natural language processing.

The attention mechanism has the advantages of plug-and-play and low computational cost. It can not only tell us where to focus but also increase the representation of the focus. Hu *et al.* [41] proposed squeeze-and-excitation networks (SENets) by utilizing and innovating feature channels in CNN. SENet learns the correlation between channels at the cost of a little more computation and achieves very good results. Woo *et al.* [42] proposed a new attention mechanism called convolutional block attention module (CBAM). It has two independent submodules, the channel attention module (CAM) and the spatial attention module (SpaAM), so that it not only pays attention to the spatial dimension but also considers the channel attention.

This article adopts the CBAM module to enhance specific features and also proposes a new SSJPA module to integrate the orientation features of the two modalities into the fused image.

III. Proposed Method

The overview architecture of the proposed FSL-Unet is shown in Fig. 2. FSL-Unet consists of two modules, including the SSAE module and the FLUF module. In the SSAE module, spectral attention and the spatial attention are computed, respectively, for the input HSI and MSI. In fact, SSAE is a module to prepare for the next stage. In FLUF, the pixel-level fusion method is adopted, and the input HSI and MSI are first fused to obtain a preliminary fused feature map. According to the large-scale range and rich detailed information of remote sensing images, more skip connections are used for full-scale multipath feature exploration. Considering the rich spectral information in HSI and the rich spatial information in MSI, we add SSJPA on the decoder side of FLUF to make the fused image preserve the spectral information of HSI and the spatial information of MSI well. On the decoder side of FLUF, feature maps of different scales are flexibly combined to gradually reconstruct HR-HSI.

A. Spatial–Spectral Attention Extraction Module

Since the spectral information of the image after the fusion of HSI and MSI mainly depends on HSI and the spatial information mainly depends on MSI, a key problem in the fusion task is how to fully transfer the information of the original image to the fused images. The attention mechanism originates from the study of human vision and can allocate efficient information processing resources to important parts. Inspired by CBAM, we extract spectral attention for HSI and spatial attention for MSI at different scales. These attentions will be used for the reinforcement of useful information and the suppression of useless information in the subsequent fusion process. Define the input HR-MSI and LR-HSI as $H \in \mathbb{R}^{H\times W\times c}$ and $L \in \mathbb{R}^{h\times w\times C}$, respectively. Fig. 3 shows the structure of the SSAE module. The specific process is given as follows.

1) Spatial Attention Extraction: First, extend the number of channels of H from c to C with a 3×3 convolution to obtain $H_1 \in \mathbb{R}^{H\times W\times C}$, which has the same spectral resolution as L. The purpose of this step is to facilitate subsequent calculations. Multiscale geometric analysis has the characteristics of multiscale and multidirection, which lays a foundation for extracting rich feature information from remote sensing images. As a member of multiscale geometric analysis, wavelet transform has the characteristics of simplicity and speed, and is valued by practical applications. Thus, in this article, we use discrete wavelet transform to downsample the feature map and inverse wavelet transformation (IWT) to upsample feature maps. In order to further extract the hierarchical information in HR-MSI, a cascade downsampling operation is performed on the H_1 to obtain $H_2 \in \mathbb{R}^{(H/2)\times(H/2)\times 2C}$ and $H_3 \in \mathbb{R}^{(H/4)\times(H/4)\times 8C}$, respectively. Next, the spatial attention of these feature maps is computed using the SpaAM. First, average pooling and max pooling operations are applied to aggregate the channel information of a feature map. The two generated 2-D maps can each represent the average pooled and max pooled features of the entire channel. Then, the two maps are concatenated and convolved to generate our spatial attention map. The computation of spatial attention at level i can be defined as

$$\mathrm{Spa}_i = \mathrm{Sig}(\mathrm{Conv}([\mathrm{AvgPool}(H_i); \mathrm{MaxPool}(H_i)])) \quad (1)$$

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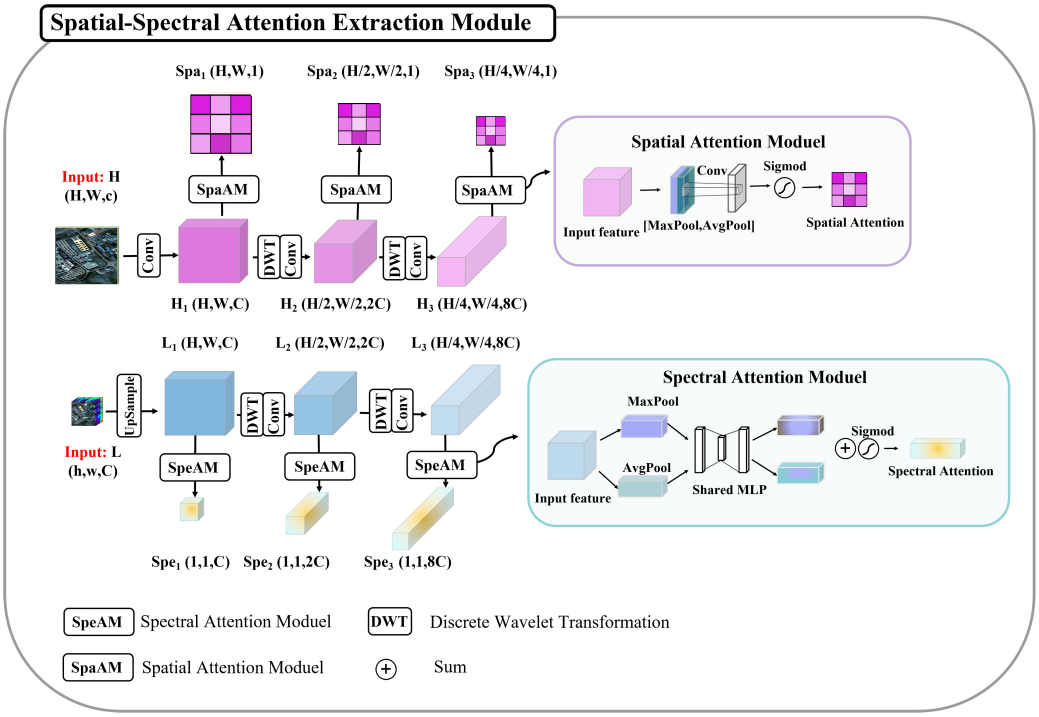


Fig. 3. Structure description of the SSAE module. SSAE can calculate the spectral attention of HSI and the spatial attention of MSI, and the calculated attention maps are applied to the FLUF module.

where Sig(·) represents the sigmoid function, Conv(·) represents the convolution operation, and [;] denotes the concatenate operation.

2) Spectral Attention Extraction: Considering the high spectral resolution of LR-HSI, we perform spectral attention extraction on LR-HSI. First, the input L is quadrupled by bilinear upsampling to obtain $L_1 \in \mathbb{R}^{H\times W\times C}$ so that L_1 has the same spatial resolution as HR-MSI. Similar to the spatial attention extraction process, L_1 is next cascaded downsampled by DWT to extract features of LR-HSI from multiple scales. We employ a spectral attention module (SpeAM) to compute the spectral attention maps of these feature maps separately. The specific calculation process is given as follows. First, the spatial information of the feature maps is integrated using two pooling operations. Two different spatial context descriptors are generated to represent average pooled features and max pooled features, respectively. These two descriptions are then fed into a shared network consisting of a multilayer perceptron (MLP) with one hidden layer to generate our final attention map. After each descriptor passes through the shared network, a summation method is used to merge the output feature vectors to obtain the final spectral attention map. The calculation process of the spatial attention map Spai at level i is

$$\mathrm{Spe}_i = \mathrm{Sig}(\mathrm{MLP}(\mathrm{AvgPool}(L_i)) + \mathrm{MLP}(\mathrm{MaxPool}(L_i))) \quad (2)$$

where Sig represents the sigmoid function and MLP denotes the multilayer perceptron.

In this way, we obtain the spatial attention maps of HR-MSI and spectral attention maps of LR-HSI at different scales, which lays the foundation for subsequent calculations. These attention maps will be applied to the encoder side of the FLUF module to realize the injection of spatial–spectral information.

B. Full-Scale Link U-Shaped Fusion Module

Fig. 4 shows the overall structure of the FLUF module. FLUF still adopts a U-shaped structure based on encoder and decoder sequences as a whole. In view of the multilevel feature extraction problem and the spatial–spectral information retention problem in the fusion process, we have made the following two innovations in U-net: 1) an SSJPA was added on the encoder side so that the feature map of each level on the encoder side can obtain the spatial information in HR-MSI and the spectral information in LR-HSI and 2) FLUF has more dense links, which make each decoder layer in FLUF contain smaller, same, and larger scale feature maps from the encoder, as well as smaller scale feature maps from the decoder, so that the hierarchical features in the network can be obtained more accurately. Next, we will introduce the above two points in detail.

1) Spatial–Spectral Joint Perceptual Attention: Similar to the human selection mechanism, the core goal of the attention mechanism in deep learning is to select the information that is more critical to the current task goal from a large amount of data and suppress useless information. The attention mechanism can not only focus on the key target area but also dedicate more attention resources to this area to obtain more detailed information and suppress other useless information. Based on the above ideas, we designed SSJPA. Therefore, unlike CBAM, SSJPA focuses more on information transfer and complementation. SSJPA adopts the mechanism of cross-attention, which can fully utilize the complementarity between input images. The fused feature map is multiplied with the spatial attention and spectral attention of the input image at the full scale, which better achieves the preservation of spatial–spectral information in the fusion process.

As shown in Fig. 5, SSJPA consists of two stages: the first stage is a spectral enhancement, and the second stage is spatial

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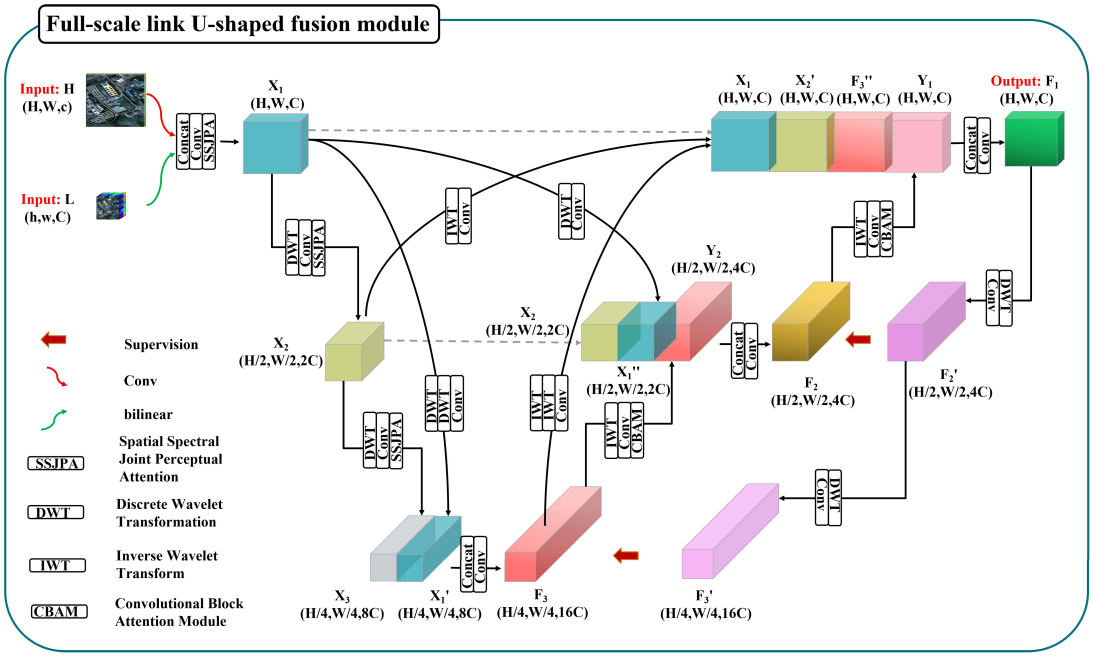


Fig. 4. Structure description of the FLUF module. FLUF consists of the encoder, decoder, and dense skip connections. FLUF can extract features in multipath and multiscale, and realize a flexible combination of features.

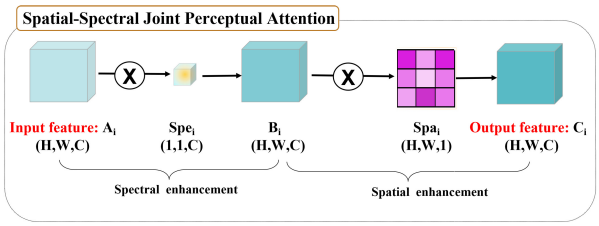


Fig. 5. Structure description of SSJPA. SSJPA consists of two stages: the first stage is spectral information enhancement, and the second stage is spatial information enhancement.

enhancement. Suppose that $A_i \in \mathbb{R}^{H\times W\times C}$ is the feature map of the ith layer on the FLUF encoder side. In the SSAE module, we obtained some spectral attention maps (Spe_i) of LR-HSI and the spatial attention maps (Spa_i) of HR-MSI at different scales. Thus, we multiply A_i by the spectral attention map and the spatial attention map at the corresponding scale, in turn, to obtain the feature map C_i with enhanced spatial–spectral information. SSJPA can be defined as

$$C_i = A_i \otimes \text{Spe}_i \otimes \text{Spa}_i \tag{3}$$

where spe_i represents the spectral attention map, spa_i represents the spatial attention map, and $\otimes$ denotes the multiplication.

2) Full-Scale Link U-Shaped Fusion: The effectiveness of super-resolution reconstruction largely depends on how to mine its high-level abstract features more effectively and how to represent the data better. Based on this idea, it is particularly important to pay attention to hierarchical features in network construction. Features at different scales have their own roles in reconstructing images. The low-level detailed feature map reflects rich spatial information, while the high-level semantic feature map reflects specific location information. Considering the stepwise downsampling and upsampling operations, the delicate signal may be gradually diluted, so dense full-scale skip connections are used in FLUF to combine multiscale features, so that the obtained spatial detail information is more abundant, and the spatial position information is more accurate.

For input images $H \in \mathbb{R}^{H\times W\times c}$ and $L \in \mathbb{R}^{h\times w\times C}$, first, spectral upsampling and spatial upsampling are performed on H and L, respectively, to obtain two feature maps with the same resolution as the fused HR-HSI. Then, the two feature maps are concatenated together and subjected to a 3×3 convolution operation to extract features. Finally, the spatial–spectral information is enhanced by the SSJPA operation, and the preliminary fusion feature map $X_1 \in \mathbb{R}^{H\times W\times C}$ is obtained. The process of this initial fusion can be defined as

$$X_1 = \mathcal{S}(\text{Conv}([\text{Conv}(H); \text{Bil}(L)])) \tag{4}$$

where $\mathcal{S}(\cdot)$ denotes the SSJPA, $\text{Conv}(\cdot)$ represents a 3×3 convolution operation, [;] is concatenate operation, and Bil represents the bilinear operation.

Furthermore, X_1 is downsampled by DWT at the encoder side. Then, perform convolution and SSJPA operations to obtain feature map X_2. Continue to repeat the above operations for X_2 to further condense the information, and obtain a feature map X_3 with a smaller spatial resolution. This process can be expressed as

$$X_n = \mathcal{S}(\text{Conv}(\text{DWT}(X_{n-1}))) \tag{5}$$

where $\mathcal{S}(\cdot)$ represents the SSJPA, $\text{Conv}(\cdot)$ represents a 3×3 convolution operation, and DWT is the discrete wavelet transformation.

Feature fusion at different scales helps greatly in pixel recovery during upsampling, so we add dense skip connections to the architecture. As shown in Fig. 4, the skip connections between different scales of the encoder and decoder are composed of IWT, DWT, and convolution. Equation (6) defines

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TABLE I
DATASET DESCRIPTION

Dataset	Pavia Center	Urban	Botswana	Pavia Uninersity	Washington DC Mall
Sensor	ROSIS	Hydice	EO-1	ROSIS	HYDICE
Location	Pavia Center, Italy.	Copperas Cove	Okavango Delta, Botswana.	Pavia Uninersity, Italy	National Mall in Washington
Image Size	1096x1096	307x307	1476x256	610x340	1280x307
SR	1.3m	2m	30m	1.3m	2.5m
Bands	102	162	145	103	191

Algorithm 1 FLUF Module

Input: Hyperspectral image L, multispectral image H, attention maps generated by the SSAM module Spa_1, Spa_2, Spa_3, Spe_1, Spe_2, Spe_3.

1: Generate initial fusion feature map X_1 via Eq.(4).
2: Shallow feature extraction and spatial spectral information enhancement to generate decoded features X_2, X_3 via Eq.(5).
3: Generate dense link path features X_1', X_1'', X_2', F_3'' via Eq. (6).
4: Flexible combination of features to generate decoded features F_3, Y_2, F_2, Y_1, F_1 via Eq.(7).

Output: Fused high-resolution hyperspectral image F_1.

these feature maps

$$\begin{cases} X_1' = \mathrm{DWT}(\mathrm{DWT}(\mathrm{Conv}(X_1))) \\ X_1'' = \mathrm{DWT}(\mathrm{Conv}(X_1)) \\ X_2' = \mathrm{IWT}(\mathrm{Conv}(X_2)) \\ F_3'' = \mathrm{IWT}(\mathrm{IWT}(\mathrm{Conv}(F3))) \\ \quad = \mathrm{IWT}\big(\mathrm{IWT}\big(\mathrm{Conv}\big(\mathrm{Conv}\big(\big[X_3; X_1'\big]\big)\big)\big)\big) \end{cases} \tag{6}$$

where Conv(·) represents a 3 × 3 convolution operation and [;] is the concatenate operation.

Finally, on the decoder side, we adaptively combine these feature maps and use CBAM to reweight the information of the feature map and gradually generate HR images through cascaded upsampling. $F_1 \in \mathbb{R}^{H\times W\times C}$ is the fused HR-HSI, and its calculation process can be described as

$$\begin{cases} F_3 = \mathrm{Conv}\big(\big[X_3; X_1'\big]\big) \\ Y_2 = \omega * (\mathrm{Conv}(\mathrm{IWT}(F_3)) \\ F_2 = \mathrm{Conv}\big(\big[\big[X_2; X_1''\big]; Y_2\big]\big) \\ Y_1 = \omega * (\mathrm{Conv}(\mathrm{IWT}(F_2)) \\ F_1 = \mathrm{Conv}\big(\big[\big[\big[X_1; X_2'\big]; F_3''\big]; Y_1\big]\big) \end{cases} \tag{7}$$

where Conv(·) represents a 3 × 3 convolution operation, [;] is the concatenate operation, and ω denotes CBAM.

Adhering to all the implementing issues discussed above, the proposed FLUF module can be summarized as in Algorithm 1.

C. Loss Function

Because super-resolution reconstruction is an ill-posed question, Guo *et al.* [43] argue that, if the mapping from LR to HR is optimal, the same input LR image can be obtained by downsampling the super-resolution image in the ideal case. Inspired by this idea, in addition to calculating the loss L_p of the fused image and the ground-truth images, FSL-Unet also downsamples fused image (F_1) to obtain F_2' and F_3', and simultaneously calculates the sum of losses (L_r) between F_2 and F_2', F_3, and F_3'. In summary, FSL-Unet consists of two parts: the original learning task and the regression learning task. Thus, primal loss L_p and regression loss L_r can be defined as

$$\begin{cases} L_p = \mathcal{L}_{\mathrm{mse}}(F_1, G) \\ L_r = \mathcal{L}_{\mathrm{mse}}\big(F_2, F_2'\big) + \mathcal{L}_{\mathrm{mse}}\big(F_3, F_3'\big) \end{cases} \tag{8}$$

where $\mathcal{L}_{\mathrm{mse}}(,)$ denotes a mean square error (mse) loss function and G is the ground-truth image.

The above operations can generate additional constraints to reduce the possible mapping space and greatly improve computational efficiency. To sum up, the total loss of FSL-Unet can be specifically defined as

$$L_{\mathrm{total}} = \lambda L_p + (1-\lambda)L_r. \tag{9}$$

Here, λ controls the weight distribution of the two-part loss, and we perform a sensitivity analysis on λ in Section IV-C1.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

In this section, we first introduce the dataset and evaluation metrics used in our experiments. Next, the experimental setup is introduced. Finally, we compare the fusion results of FSL-Unet on five hyperspectral datasets with eight state-of-the-art algorithms, and the ablation experiments of different modules in the model are carried out to verify its effectiveness.

A. Dataset Description

In this article, we use five publicly available hyperspectral datasets to verify the effectiveness of the FSL-Unet, including Pavia Center, Urban, Botswana, Pavia University, and Washington DC Mall. Table I reports the detail information of the five datasets.

1) Pavia Center: Pavia Center data are acquired by ROSIS sensors. The size of Pavia Center is 1096 × 1096, and there are 115 bands in total. After processing, this article uses the remaining 102 bands.

2) Urban: The Urban dataset was acquired by the Hydice sensor in 1995. It has a spatial resolution of 2.5 m and a size of 307 × 307. The Urban data consist of 210 bands, and after removing noise and water absorption bands, this article utilizes the remaining 162 bands.

3) Botswana: The Botswana dataset was acquired by the EO-1 satellite in Botswana from 2001 to 2004. It has 242 bands and a spatial resolution of 30 m. After removing the noise band, this article uses the remaining 145 bands.

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TABLE II

SENSITIVITY ANALYSIS RESULTS FOR PARAMETER λ ON THE URBAN DATASET. THE BEST AND SECOND SCORES ARE MARKED IN **RED** AND **BLUE**, RESPECTIVELY

λ	Urban			
	RMSE	PSNR	ERGAS	SAM
0	56.3835	10.8601	3.9397	57.1667
0.1	5.2889	31.4142	3.3738	4.9405
0.2	2.5210	37.8518	1.2704	2.3258
0.3	**2.4623**	**38.0562**	1.3036	2.3258
0.4	2.4678	38.0369	**1.2611**	**2.2603**
0.5	2.6134	37.5435	1.3636	2.4566
0.6	**2.3706**	**38.3858**	**1.1947**	**2.1956**
0.7	2.5832	37.6400	1.3111	2.3480
0.8	2.5501	37.7519	1.2847	2.3336
0.9	2.4662	38.0425	1.2552	2.2687
1.0	2.5149	37.8726	1.3212	2.3462

4) Pavia University: The Pavia University dataset was captured by the ROSIS-03 sensor in 2003. It has a spatial resolution of 1.3 m and a size of 610 × 340. Pavia University has 115 bands, 12 bands were removed because of noise influence, and the remaining 103 bands are used.

5) Washington DC Mall: The Washington DC Mall dataset is an aerial HSI acquired by the Hydice sensor in 1995. It consists of 210 bands, with a spatial resolution of 2.5 m and a size of 1280 × 307. Excluding the 19 bands affected by water vapor, the remaining 191 bands are used in this article.

In the experiments, we generated two input images with reference images according to Wald's protocol [44]. Define L as the input LR-HSI, which is obtained by Gaussian blurring and quadruple downsampling of the reference image. Define H as the input HR-MSI, which is formed by extracting five bands from the reference image at equal intervals. We use a 128 × 128 pixel subregion of the data center for testing and the rest for training. The training region is randomly cropped to a size of 128 × 128 pixels for each iteration of training. In order to avoid the regions overlapping between training and testing samples, we apply zero-padding to the test regions of the image samples during model training.

B. Evaluation Metrics

This article uses four evaluation metrics to measure the quality of fused images, including the root mean square error (RMSE), the peak signal-to-noise ratio (PSNR), the Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS), and SAM. The significance of these four evaluation metrics will be introduced in detail next. Here, we define $F_k(i, j)$ to represent the reference HR-HSI at the position of (i, j) at the K-band, and $\hat{F}_k(i, j)$ represents the estimated HR-HSI at the K-band.

1) Root Mean Square Error: RMSE is a measure of the deviation between the estimated data and the ground truth, which can be defined as

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{k=1}^{C}\sum_{i=1}^{H}\sum_{j=1}^{W}\left(F_k(i,j)-\hat{F}_k(i,j)\right)^2}{HWC}} \tag{10}$$

where H, W, and C represent the height, the width, and the number of channels of the image, respectively. The smaller the RMSE, the better the quality of the fused image.

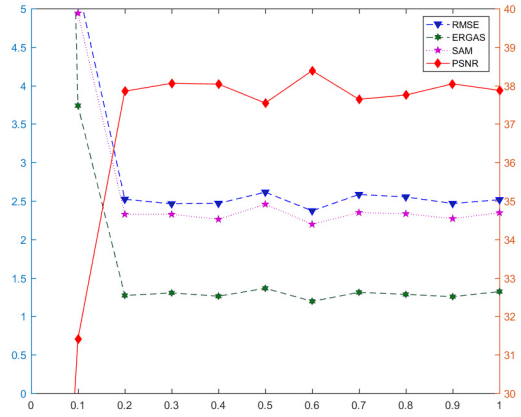


Fig. 6. Change trend of each indicator under different λ values on the Urban dataset. The values of RMSE, REGAS, and SAM are referenced to the left y-axis coordinate value, and PSNR is referenced to the right y-axis value.

2) Peak Signal-to-Noise Ratio: PSNR is currently the most widely used and the most common method for objectively measuring image quality. The PSNR of the ith band is defined as

$$\mathrm{PSNR} = 10\log_{10}\left(\frac{\max(F_i)^2}{\frac{1}{HW}\left\|F_i-\hat{F}_i\right\|_2^2}\right) \tag{11}$$

where H and W represent the height and width of the image, respectively. $\|\cdot\|_2$ is the two-norm.

3) Erreur Relative Globale Adimensionnelle de Synthèse: ERGAS can measure the similarity between the estimated image and the ground-truth image. The smaller the value of ERGAS, the better the performance of the model. It is defined as

$$\mathrm{ERGAS} = 100\frac{h}{l}\sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\mathrm{RMSE}^2(B_i)/M_i^2\right)} \tag{12}$$

where h is the resolution of HR-MSI, l is the resolution of LR-HSI, N is the number of bands, B_i is HR-MSI, and M_i is the average of the emissivity of the HR-MSI.

4) Spectral Angle Mapper (SAM): SAM is used to calculate the similarity between two spectral curves, which can effectively measure the difference between two remote sensing images. SAM can be defined as

$$\mathrm{SAM} = \arccos\left(\frac{\langle F(i,j),\hat{F}(i,j)\rangle}{\|F(i,j)\|_2\|\hat{F}(i,j)\|_2}\right) \tag{13}$$

where $\langle\cdot,\cdot\rangle$ represents the inner product operation. Average the SAM values of all pixels to get the final SAM result.

C. Parameter Tuning

The operating system used in this experiment is 64-bit Windows 10 and NVIDIA Quadro P6000 GPU. The deep learning framework used is Pytorch 1.2.0, and the programming language is Python 3.7. The Adam optimizer was used in the training process.

The setting of hyperparameters in deep learning networks often plays a crucial role in the performance of the model, so we conduct experiments on the settings of parameter λ and

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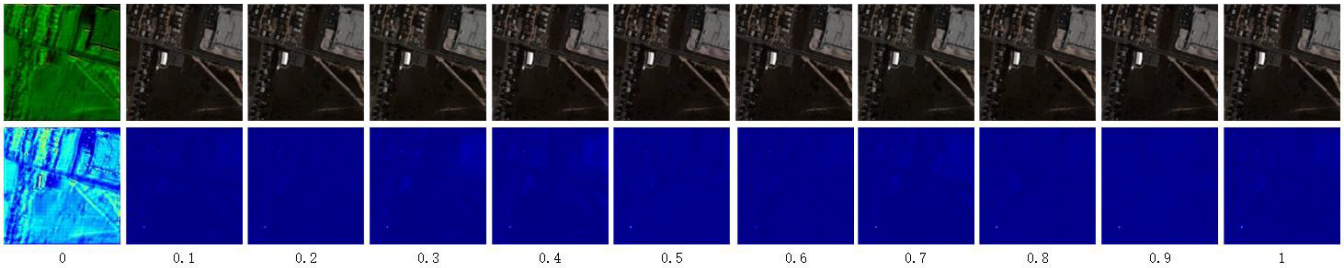


Fig. 7. Fusion results of different λ's on the Urban dataset. The first row shows the pseudocolor RGB result images after fusion, and the second row shows the difference images after pseudocolor processing. The numbers below each column represent the value of λ.

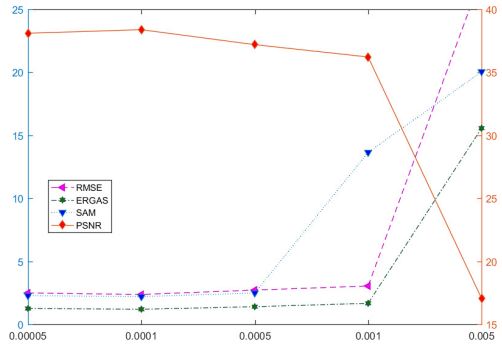


Fig. 8. Model performance comparison under different learning rates. The values of RMSE, ERGAS, and SAM are referenced to the left y-axis coordinate value, and PSNR is referenced to the right y-axis value.

learning rate, and analyze and discuss the impact on the model reconstruction ability.

1) Sensitivity Analysis of λ: As mentioned in Section III-C, the loss function of our model consists of two parts. In (9), L_p is the constraint for fused image and ground truth, and L_r is an additional constraint on the network, which is used to reduce possible mappings and improve learning efficiency. λ controls the weight distribution of the two-part loss. To discuss the effect of λ on network performance, we set λ from 0 to 1 with an interval of 0.1. Table II shows the results of RMSE, PSNR, ERGAS, and SAM of fused images on the Urban dataset. Fig. 6 shows the changing trend of each indicator under different λ values. Fig. 7 shows the fused images corresponding to different λ's. When λ is 0, the performance of the model is very poor. With the continuous increase of λ, each index gradually achieves better results, and it is the best when it is 0.6. It is worth mentioning that, when λ is 1, that is, when only L_p is used as the loss function, the model performance is average, which fully demonstrates the effectiveness of L_p and L_r in improving model performance. In summary, for other experiments in this article, we uniformly set λ to 0.6.

2) Learning Rate Comparison: The learning rate is an important parameter of the deep learning framework, and choosing an appropriate learning rate can speed up the convergence of the objective function. To verify the effect of different learning rates on model performance, we conduct experiments on the urban dataset, and Fig. 8 shows the line graph of the indicator results under different learning rates. The experimental results show that the model performance first gets better and then worse with the increase of the learning rate. When the learning rate is $1e-4$, the model performance is optimal.

TABLE III

COMPARATIVE EXPERIMENTAL RESULTS ON THE PAVIA CENTER DATASET. THE BEST AND SECOND SCORES ARE MARKED IN **RED** AND **BLUE**, RESPECTIVELY

Method	Pavia Center			
	RMSE	PSNR	ERGAS	SAM
CNMF	13.9777	25.2221	11.4360	4.3635
LTTR	28.3373	19.0836	31.1229	17.0800
MSDCNN	4.1687	35.7370	4.6840	5.4004
TFNet	4.2473	35.5686	4.8295	4.8437
ResTFNet	3.9258	36.2522	4.4449	4.6073
SSFCNN	4.7720	34.5569	5.7509	5.7962
ConSSFCNN	5.4998	33.3239	6.1594	6.1196
SSR-NET	**3.4102**	**37.4752**	**3.8964**	**3.8618**
FSL-Unet	**2.6718**	**39.5946**	**3.0431**	**3.4794**

D. Comparison With State-of-the-Art Methods

To verify the excellent performance of the proposed FSL-Unet, we conduct comparative experiments with eight state-of-the-art methods on five publicly available hyperspectral datasets. There are two traditional methods and six deep learning methods for comparative experiments, including CNMF [26], LTTR [45], TFNet [36], ResTFNet [36], SSFCNN [35], ConSSFCNN [35], MSDCNN [34], and SSR-NET [37]. We use the form of a table to display the objective evaluation indicators of the fused image and use the form of a subjective map to display the fused image. Due to space constraints, we only show the result plots of the deep-learning-based fusion method. To be more intuitive and clear, we not only show the pseudocolor RGB image of the estimated image but also show the difference image between the estimated image and the reference image. The following is an analysis of the experimental results on each dataset.

1) Results on Pavia Center: Table III and Fig. 9 show the evaluation metrics and pseudocolor RGB images of the reconstruction results on the Pavia dataset, respectively. Compared with SSR-NET, the second-ranked reconstruction algorithm, FSL-Net, improves PSNR by 5.65% and reduces RMSE, ERGAS, and SAM by 21.65%, 21.90%, and 9.90%, respectively. These data fully illustrate the effectiveness of FSL-Net on the super-resolution reconstruction task of the Pavia Center dataset. Especially for ERGAS, a considerable reduction is achieved, which fully demonstrates that FSL-Net has excellent spatial and spectral information retention capabilities.

2) Results on Urban: Table IV and Fig. 10 show the evaluation metrics and pseudocolor RGB images of the reconstruction results on the Urban dataset, respectively. Experimental results show that FSL-Net achieves competitive reconstruction

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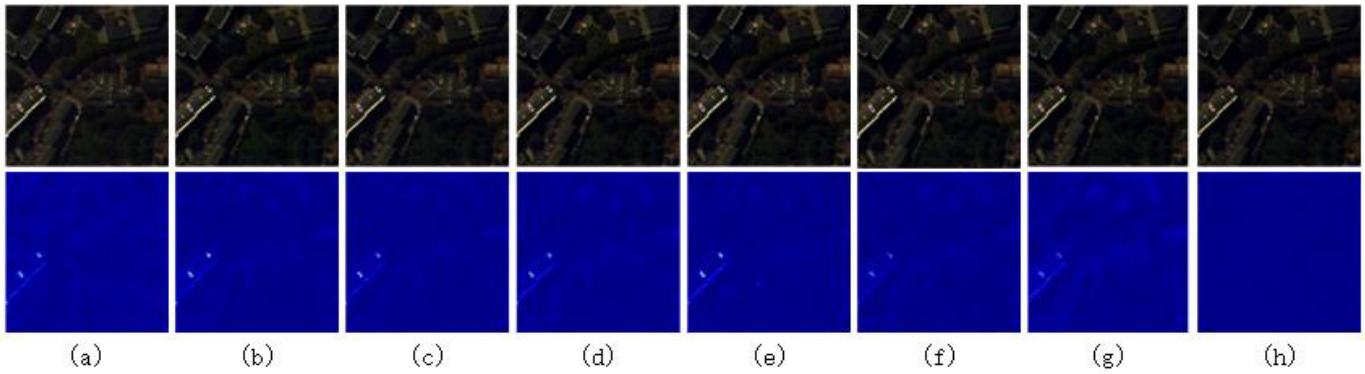


Fig. 9. Fusion results of different methods on the Pavia Center dataset. The first row shows the pseudocolor RGB result images after fusion, and the second row shows the difference images after pseudocolor processing. (a) MSDCNN. (b) TFNet. (c) ResTFNet. (d) SSFCNN. (e) ConSSFCNN. (f) SSR-NET. (g) FSL-Unet. (h) Ground truth.

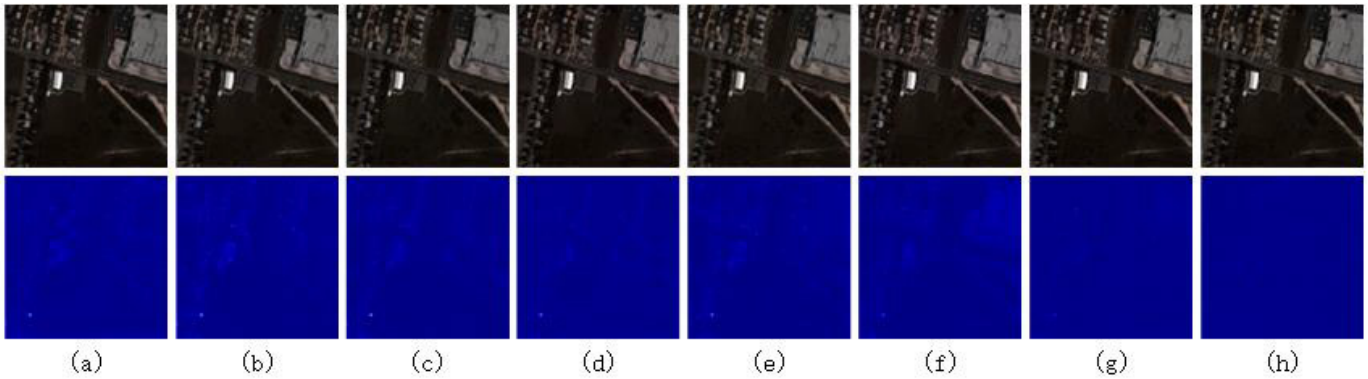


Fig. 10. Fusion results of different methods on the Urban dataset. The first row shows the pseudocolor R-G-B result images after fusion, and the second row shows the difference images after pseudocolor processing. (a) MSDCNN. (b) TFNet. (c) ResTFNet. (d) SSFCNN. (e) ConSSFCNN. (f) SSR-NET. (g) FSL-Unet. (h) Ground truth.

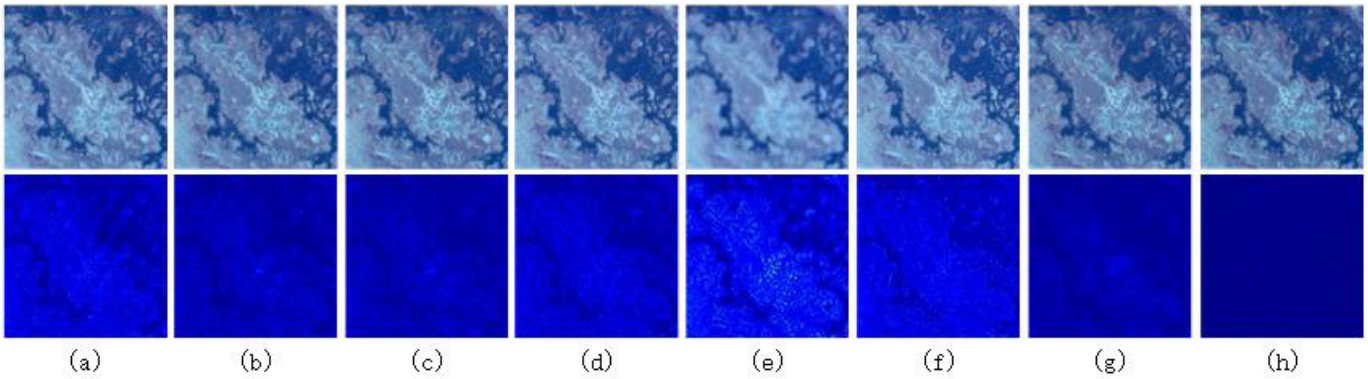


Fig. 11. Fusion results of different methods on the Botswana dataset. The first row shows the pseudocolor R-G-B result images after fusion, and the second row shows the difference images after pseudocolor processing. (a) MSDCNN. (b) TFNet. (c) ResTFNet. (d) SSFCNN. (e) ConSSFCNN. (f) SSR-NET. (g) FSL-Unet. (h) Ground truth.

results on the Urban dataset. Among them, SSR-NET won first place in the RMSE and PSNR metrics, which mainly benefited from the spatial loss function. FSL-Net achieves the best performance in ERGAS and SAM, once again proving its excellent spectral preservation ability. Thanks to SSJPA in the network, the spectral information in the HSI is transferred to the fusion image during the fusion process, which greatly avoids the loss of spectral information.

3) Results on Botswana: Table V shows the experimental results on the Botswana dataset; Fig. 11 shows the subjective results on the Botswana dataset and the difference images between the fusion results and the reference HR-HSIs. The results on the Botswana dataset are similar to those on the Urban dataset, where FSL-Unet achieves two best and two competitive metrics results. It is worth mentioning that the ERGAS of the fused image of FSL-Unet is 30% lower than that of SSR-NET, which is a great improvement. It is demonstrated that the dense links in FSL-Unet can fully utilize the features of various scales, which is crucial for the recovery of spatial details in the reconstruction process.

4) Results on Pavia University: Table VI shows the objective results of all fusion methods, and Fig. 12 shows the fused images and the difference image between them and the ground truth. Experimental results show that, compared

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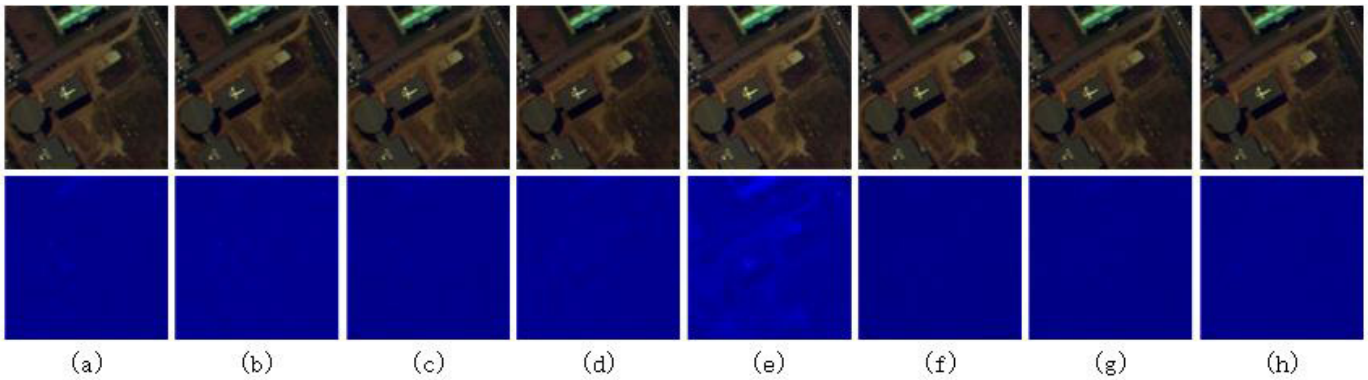


Fig. 12. Fusion results of different methods on the Pavia University dataset. The first row shows the pseudocolor R-G-B result images after fusion, and the second row shows the difference images after pseudocolor processing. (a) MSDCNN. (b) TFNet. (c) ResTFNet. (d) SSFCNN. (e) ConSSFCNN. (f) SSR-NET. (g) FSL-Unet. (h) Ground truth.

TABLE IV

COMPARATIVE EXPERIMENTAL RESULTS ON THE URBAN DATASET. THE BEST AND SECOND SCORES ARE MARKED IN **RED** AND **BLUE**, RESPECTIVELY

Method	Urban			
	RMSE	PSNR	ERGAS	SAM
CNMF	4.0198	36.0468	1.5612	2.4971
LTTR	24.6837	20.2826	11.7672	9.1336
MSDCNN	3.1317	35.9676	1.7652	3.1199
TFNet	3.0405	36.2243	1.7146	2.8524
ResTFNet	2.8916	36.6604	1.5980	2.7355
SSFCNN	8.5801	27.2133	4.2196	8.7402
ConSSFCNN	3.8841	34.0972	1.8671	3.1609
SSR-NET	**2.3693**	**38.3909**	**1.2173**	**2.3142**
FSL-Unet	**2.3706**	**38.3858**	**1.1947**	**2.1956**

TABLE V

COMPARATIVE EXPERIMENTAL RESULTS ON THE BOTSWANA DATASET. THE BEST AND SECOND SCORES ARE MARKED IN **RED** AND **BLUE**, RESPECTIVELY

Method	Botswana			
	RMSE	PSNR	ERGAS	SAM
CNMF	26.3457	19.7166	9.4849	2.4866
LTTR	9.1525	28.9000	14.2396	4.8682
MSDCNN	0.5544	36.3497	3.1753	2.5996
TFNet	0.4991	37.2622	2.7884	2.3161
ResTFNet	0.4598	37.9749	2.6952	2.1537
SSFCNN	1.4151	28.2108	11.4498	7.0424
ConSSFCNN	2.5058	23.2475	15.2048	12.9631
SSR-NET	**0.4106**	**38.9583**	**2.7994**	**2.0181**
FSL-Unet	**0.4110**	**38.9505**	**1.9596**	**1.8909**

TABLE VI

COMPARATIVE EXPERIMENTAL RESULTS ON THE PAVIA UNIVERSITY DATASET. THE BEST AND SECOND SCORES ARE MARKED IN **RED** AND **BLUE**, RESPECTIVELY

Method	Pavia University			
	RMSE	PSNR	ERGAS	SAM
CNMF	7.5816	30.5356	3.8939	2.2469
LTTR	9.4202	28.6469	5.5504	5.1883
MSDCNN	2.3207	40.5032	1.6358	2.6309
TFNet	2.2427	40.8004	1.6053	2.4890
ResTFNet	2.0578	41.5477	1.5068	2.3517
SSFCNN	1.9410	42.0550	1.3957	2.2121
ConSSFCNN	2.3916	40.2420	1.6340	2.4235
SSR-NET	**1.6447**	**43.4938**	**1.2146**	**1.9329**
FSL-Unet	**1.4802**	**44.3987**	**1.1287**	**1.7944**

TABLE VII

COMPARATIVE EXPERIMENTAL RESULTS ON THE WASHINGTON DC MALL DATASET. THE BEST AND SECOND SCORES ARE MARKED IN **RED** AND **BLUE**, RESPECTIVELY

Method	Washington DC Mall			
	RMSE	PSNR	ERGAS	SAM
CNMF	57.7621	12.8978	1356.2909	32.6869
LTTR	18.6020	22.7396	301.2300	13.3932
MSDCNN	2.7328	36.7345	0.4778	0.8993
TFNet	1.9704	39.5756	0.3460	0.6756
ResTFNet	**1.7348**	**40.6816**	**0.3035**	**0.5836**
SSFCNN	16.4613	21.1373	3.1689	6.8978
ConSSFCNN	14.0913	22.4876	2.6805	5.8734
SSR-NET	2.0591	39.1928	0.3603	0.7201
FSL-Unet	**0.7023**	**48.5363**	**0.1216**	**0.2415**

with other reconstruction algorithms, FSL-Unet achieves the best performance on all four metrics. The fused images have a certain degree of improvement in both spatial quality and spectral quality, which shows that FSL-Unet not only has the ability to preserve the spectrum but also can make full use of the spatial information of MSI to improve the quality of the fused images.

5) Results on Washington DC Mall: The evaluation index results of each fusion method on the Urban dataset are shown in Table VII; Fig. 13 shows the subjective results and the difference image between them and the ground truth. FSL-Unet achieves the best results on all four metrics on the Washington DC Mall dataset. Compared with ResTFNet, the second-ranked method in reconstruction quality, FSL-Net, improves PSNR by 59.52% and reduces RMSE, ERGAS, and SAM by 16.80%, 59.93%, and 58.62%, respectively. It can be seen that FSL-Unet has achieved a huge leap in performance on the Washington DC Mall dataset. Compared with other datasets, the Washington DC Mall dataset has more bands, indicating that FSL-Unet has excellent performance for high spectral resolution datasets.

E. Ablation Study

To verify the influence of each module in FSL-Unet on the reconstruction results, we conduct ablation experiments on the Urban dataset.

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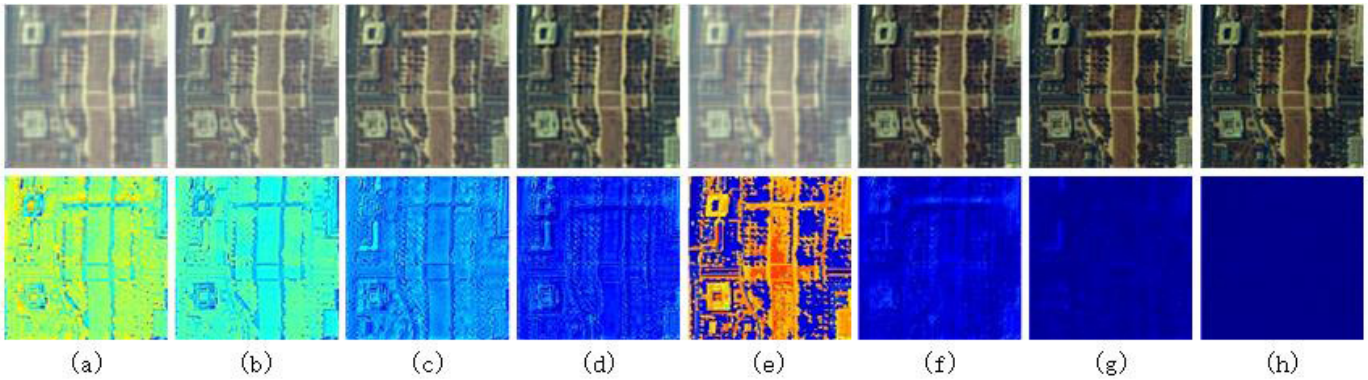


Fig. 13. Fusion results of different methods on the Washington DC Mall dataset. The first row shows the pseudocolor R-G-B result images after fusion, and the second row shows the difference images after pseudocolor processing. (a) MSDCNN. (b) TFNet. (c) ResTFNet. (d) SSFCNN. (e) ConSSFCNN. (f) SSR-NET. (g) FSL-Unet. (h) Ground truth.

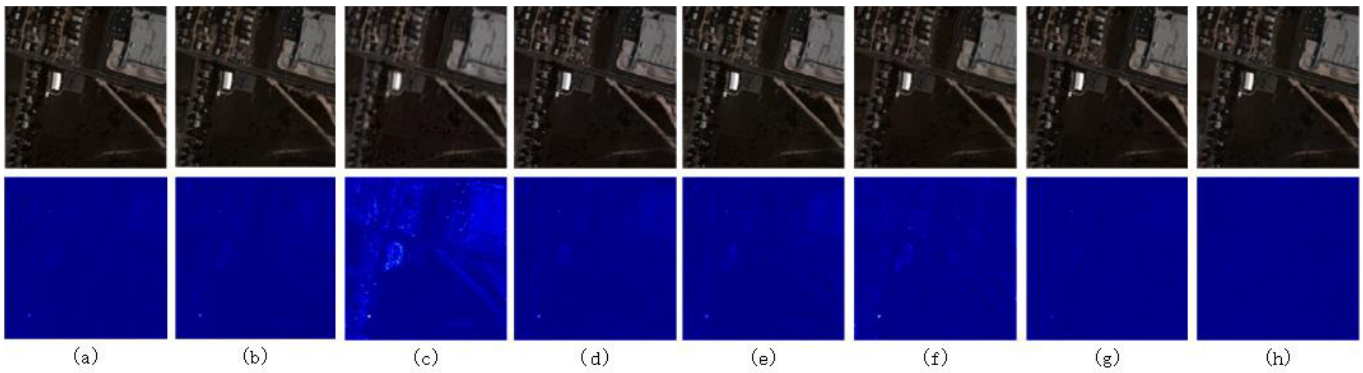


Fig. 14. Fusion results of Ablation study on SSJPA and full-scale link on the Urban dataset. The first row shows the R-G-B result images after fusion, and the second row shows the difference images after pseudocolor processing. (a) Removing spectral enhancement. (b) Removing spatial enhancement. (c) Removing the entire SSJPA. (d) Repiacing SSJPA with CBAM. (e) Repiacing CBAM with SSJPA. (f) Removing full-scale link. (g) FSL-Unet. (h) Ground truth.

1) Ablation Study on SSJPA: SSJPA is used at the decoder side of FSL-UNet to transfer the rich spatial information in HR-MSI and the rich spectral information in LR-HSI into the fused image at full scale. The calculation process of SSJPA can be divided into two stages: spectral enhancement and spatial enhancement. To verify the effectiveness of SSJPA, we conduct five ablation experiments: 1) replacing SSJPA in the decoder side with CBAM; 2) replacing CBAM in the encoder side with SSJPA; 3) removing the spectral enhancement in SSJPA; 4) removing the spatial enhancement in SSJPA; and 5) removing the entire SSJPA.

Table VIII and Fig. 14 show the objective evaluation index results of the experiment and the subjective results of the fused images, respectively. The results show that the model performance is degraded regardless of replacing SSJPA on the decoder side with CBAM or replacing CBAM with SSJPA. SSJPA enables the feature maps of each level in the encoder side to obtain spatial information of the MSI and spectral information of the HSI, which is conducive to the preservation of spatial–spectral information in the fusion process. The decoder-side CBAM is to utilized to further highlight the representative spatial–spectral integrated features. The results of ablation experiments demonstrate the effectiveness of this design. The results also show that, when the spectral enhancement stage is removed, the quality of the fused image decreases slightly, which indicates that the spectral enhancement makes effects on the preservation of spectral and spatial information. When the spatial enhancement stage is removed, the quality of the fused images deteriorates significantly. The phenomenon shows that the interaction of spatial and spectral information with the fused image plays a crucial role. When all SSJPA is removed, the model suffers a serious performance decreasing. Our proposed SSJPA is able to propagate key information to the fused images during the fusion process, which casts the model the better representation ability.

TABLE VIII

RESULTS OF ABLATION STUDY ON SSJPA AND FULL-SCALE LINK ON THE URBAN DATASET. THE BEST SCORES ARE MARKED IN **RED**

Architecture	Urban			
	RMSE	PSNR	ERGAS	SAM
Removing spectral enhancement	2.6411	37.4475	1.3314	2.3526
Removing spatial enhancement	6.7597	29.2846	3.3955	6.7436
Removing the entire SSJPA	7.8399	27.9970	4.1995	7.1049
Replacing SSJPA with CBAM	2.5334	37.8092	1.2888	2.2909
Replacing CBAM with SSJPA	2.5746	37.6690	1.3084	2.3820
Without Full-scal link	3.9136	34.0316	2.6637	3.7040
FSL-Unet	**2.3706**	**38.3858**	**1.1947**	**2.1956**

2) Ablation Study on Full-Scale Link: The full-scale dense links in FSL-Unet can realize multipath acquisition and fusion of features of different scales. This flexible and adaptive feature fusion scheme greatly improves the representation ability of the network. We remove dense links in the network to verify their impact on model performance. Table VIII and Fig. 14

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TABLE IX
TEST TIME IN MICROSECOND OF THE TESTING METHODS

Method	MSDCNN	TFNet	ResTFNet	SSFCNN	ConSSFCNN	SSR-Net	FSL-UNet
Pavia Center	141	135	151	135	143	144	210
Urban	140	145	145	134	132	140	256
Botswana	146	147	136	137	135	127	253
pavia University	138	140	142	138	136	131	209
Washington DC	148	147	138	129	134	130	290

show the objective evaluation metrics of the experiment and the subjective results of the fused images, respectively. The experimental results show that, when the full-scale links are removed from the network, the performance of the model is greatly reduced, and the quality of the fused image has deteriorated. It shows that the full-scale link can extract information through multiple paths, realize the effective aggregation of features, and effectively improve the representation ability of the network, which is very important for super-resolution tasks.

F. Time Efficiency Analysis of Model

To compare the time efficiency of the test methods, Table IX reports the test time of the different methods on each dataset. All experiments were performed on a Win10 computer equipped with an NVIDIA Quadro P6000 GPU. Since FSL-Unet has many attention mechanisms and dense links, these can reduce computational efficiency while enabling feature reuse and preserving spatial–spectral information. As shown in Table VII, FSL-Unet archives the best performance with a test time of 290 ms, which achieves a balance between time efficiency and reconstruction image quality.

V. CONCLUSION

This article proposes a novel CNN-based hyperspectral and MSI fusion network, called FSL-Unet. FSL-Unet consists of two components: the SSAE module and the FLUF module. Due to the high spatial resolution of HR-MSI and the high spectral resolution of LR-HSI, the SSAE module makes full use of the image properties to extract spatial attention and spectral attention for HR-MSI and LR-HSI, respectively. In the FLUF module, fully considering the complementarity of HR-MSI and LR-HSI in the spatial mode and the spectral mode, we design an SSJPA applied to the encoder side to make the spatial information and spectral information effective embed in the fused image to complete the interaction and integration of information across modalities. The FLUF module adopts a multilevel feature fusion strategy, and the full-scale link realizes the acquisition and adaptive combination of multipath and multiscale features. Experimental results on five widely used hyperspectral datasets demonstrate the effectiveness of FSL-Unet.

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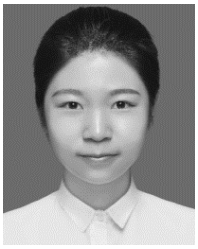
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